

Programming using Python (CSIC 104)
Mid Sem Exam-2
Robotics and Automation

Total Marks- 15

1. **OOP and exception handling –Short questions (7 Marks)**
 - a) Create a class BankAccount with attributes account_holder and balance. Implement methods for deposit() and withdraw(). Ensure withdraw() doesn't allow overdrafts. (Explain with an example) (2 Marks)
 - b) Create a subclass SavingsAccount of a base class BankAccount that adds an attribute interest_rate and a method calculate_interest(). Demonstrate the concept of method overriding. (2 Marks)
 - c) Explain the difference between multiple and multilevel inheritance with an example. (1 Marks)
 - d) What is the difference between class variables and instance variables? How to use those variables in main program. Explain with an example. (1 Marks)
 - e) What is the importance of raising custom exceptions? Explain with an example. (1 Mark)
2. **Turtle Graphics (3 Marks)**
 - a) Write a Python program that allows the user to enter the number of sides, and then turtle should draw a regular polygon with that many sides. (1.5 Marks)
 - b) Write a Python program to draw the image shown with different color. (1.5 Marks)
3. Design a **University Management System** using Python OOP, incorporating **Student, Teacher, and Admin** classes. The system should handle **student admissions, course enrollment, grade allotment, student ranking, and staffing** while utilizing **inheritance and dictionaries** for data management. Implement a Python program demonstrating these functionalities. Explain with an example. (5 marks)

